

Control room

Revise 360 lesson plan — 60 minutes

Unit	OCR J277 1.5 Systems software — Paper 1, Computer systems
Lesson	Control room (lesson 1 of the unit)
Lesson ID	ss-l01
Experience ID	ss-l01
Plan ID	PLAN-SS-L01
Content version	1bdd3d93c7f2
Website	https://revise360.co.uk/experience.html?id=ss-l01

Big question, outcomes and success criteria

Big question — What does an operating system do, and why does every computer need one?

Prerequisites

None within this unit: it is the first lesson. Treat the starter as a baseline check rather than retrieval of something you have taught.

Assessable outcomes

1. **ss-l01-o1** — know the purpose and functionality of operating systems
2. **ss-l01-o2** — know the different types of user interface and the features of each

Vocabulary, resources and preparation

Key vocabulary — Operating system, User interface, Control room

Resources

SLIDES	presentations/SS_L01_ControlRoom_Lesson.pptx
WORKSHEET	worksheets/SS_L01_ControlRoom_Worksheet.docx
ONLINE EXPERIENCE	experience.html?id=ss-l01

Before the lesson

1. Sign in on the **teacher dashboard** (teacher.html) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
4. Check headsets or devices are charged and on the school network.

How pupils get in

1. Go to `topics.html` on `revise360.co.uk`.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.
3. Choose the topic, then this lesson's card.

Worth knowing before you promise anything. In this build APP_CONFIG.backendUrl is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
0-5	Retrieve	Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites.	Answer the starter on the worksheet.	The starter box on SS_L01_ControlRoom_Worksheet.docx is filled in.
5-8	Review the starter	Take answers; correct the one most got wrong.	Correct their own starter.	A correction written, not just a tick.
8-18	Explain and model	The new idea in two or three steps, with the example worked in front of the class.	Listen and answer questions about the example.	They can say why, not only what.
18-21	Set up and settle	Sign-in, launch, and what to write before anyone taps an answer.	Sign in, open the lesson, find What is systems software?.	Everyone is in and on station 1.
21-43	Online: work the stations in order	Pupils read each wall, write the key fact and the challenge, then answer at the badge.	Read each wall, write the key fact and the challenge, then answer at the badge.	The worksheet has writing on it for each of the 6 stations before the badge is tapped.
43-48	Check understanding	The hinge question, hands down.	Answer the hinge question, hands down.	The split across the options.
48-56	Apply	Exam practice on the worksheet, in silence.	Exam practice, in silence.	worksheet, Exam practice a) to c)
56-60	Exit and set up the next lesson	The exit question and the retrieval link to next lesson.	Write the exit answer.	One sentence that answers the big question.

The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

1. What is systems software?

- Software is split into two kinds: systems software and application software.
- Systems software runs the computer itself: the operating system and utility programs.
- Application software is what the user runs to do a job: a browser, a game, a word processor.

Challenge on this wall: Write your own definition of systems software, with two examples.

2. What the OS does

- It manages the processor, sharing it between programs.
- It manages memory, deciding what is held where.
- It manages files, users, and input and output devices.
- It provides a user interface so people can work with the machine.

Challenge on this wall: List the five jobs an operating system does, in your own words.

3. Graphical user interface

- A GUI uses windows, icons, menus and a pointer.
- It is easy to learn, and needs no commands to be memorised.
- It uses more memory and processing power than a text interface.

Challenge on this wall: Give one advantage and one disadvantage of a GUI.

4. Command line interface

- A CLI is text only: the user types commands.
- It uses very little memory and processing power.
- It is powerful and easy to automate with scripts, but commands must be learned.

Challenge on this wall: Give one situation where a CLI is a better choice than a GUI.

5. Menu driven and voice

- A menu driven interface offers a set of choices, one screen at a time: a cash machine, a ticket machine, an old MP3 player.
- It is very easy to use, but slow, and you can only do what the menus allow.
- A voice interface listens for spoken commands: useful hands free, and for users who can't use a screen, but it mishears and needs a quiet room.

Challenge on this wall: Choose an interface for a cash machine, a games console and a server, and justify each choice.

6. Choosing an interface

- Think about the user: their skill, their situation and what they need to do.
- Think about the hardware: a device with no screen can't use a GUI.
- Exam answers need the choice, the characteristic and the reason.

Challenge on this wall: Explain which interface you would choose for a self-service checkout, and why.

Retrieval and the check for understanding

Starter — How many operating systems can you name? For each one, say what kind of device it runs on.

Answer space: 2 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (ss-101-s4-c1), so the wording a pupil sees matches the wording you say.

Why do network managers often use a command line?

Correct: Commands can be scripted, so the same job runs on hundreds of machines

What a pupil might choose instead

- It looks more impressive
- It uses more memory
- It has bigger icons

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

The course's own response to a wrong answer here: One script can do in seconds what would take hours of clicking.

One response, not four. one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

Guided to independent practice

STAGE	ACTIVITIES	WHAT IT IS
Guided	13	Support is in the activity: an example to run, an output to predict, one thing to change, a gap to fill.
Independent	0	The pupil writes or fixes it with the brief and nothing else.
Challenge	0	The hardest activity on each station. It is not optional and it is not extra: it is the last step of the station.

Every activity in this lesson is guided: a pupil chooses between options and never produces anything of their own. Plan the independent work yourself - the exam practice on the worksheet is the obvious place - and do not read a full score here as evidence that a pupil can do it unaided.

Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

Delivering the online experience

Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

The stations, and what to record at each

#	STATION	ACTIVITIES	RECORD ON PAPER
1	What is systems software?	2	Systems software is...
2	What the OS does	2	An operating system manages...
3	Graphical user interface	2	A graphical user interface...
4	Command line interface	2	A command line interface...
5	Menu driven and voice	2	A menu driven interface... A voice interface...

#	STATION	ACTIVITIES	RECORD ON PAPER
6	Choosing an interface	2	To choose an interface, you...
final	Final challenge: pick the interface	1	the answer to the station's challenge

Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

Assessment, evidence and unfinished work

ID	QUESTION	MARKS	WHERE
ss-101-x1	State what is meant by systems software, and give two examples.	3	worksheet, Exam practice a); deck, Exam practice slide
ss-101-x2	Describe two jobs carried out by an operating system.	4	worksheet, Exam practice b); deck, Exam practice slide
ss-101-x3	A museum is building an information point for visitors. Recommend a type of user interface and justify your choice.	4	worksheet, Exam practice c); deck, Exam practice slide

Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/ss-101.json`, which is not published with the website.

Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **Which of these is systems software?** — A web browser; A spreadsheet
- **Why does an application not talk to the hardware directly?** — Hardware can't be programmed; It would be too fast
- **What is the main advantage of a GUI?** — It uses less memory; It is faster for experts to automate
- **What is the main disadvantage of a GUI?** — It can't show pictures; It needs a keyboard
- **Why do network managers often use a command line?** — It looks more impressive; It uses more memory

Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

Exit, next lesson and homework

Exit check

One of these, written, not a show of hands:

- Systems and application software
- What an operating system does
- The four types of user interface

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

Retrieval link to next lesson

Next is ss-102, Multitasking floor. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

Shorter lessons, longer lessons and no internet

A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

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